



Original Research

KGiA: Drug repurposing through disease-aware knowledge graph augmentation

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ABSTRACT

Objective: Drug repurposing offers a cost-effective strategy to accelerate drug development by identifying new therapeutic uses for approved medications. Knowledge graphs (KGs) that capture large amounts of biomedical knowledge have recently been used for drug repurposing, however, KGs are inherently incomplete due to our limited biomedical knowledge.

Methods: We propose KGiA, an inductive graph augmentation method that supports semi-inductive reasoning—allowing models to generalize to previously unseen biomedical entities. KGiA enhances KGs using counterfactual relationships mined from disease-specific topological patterns. We apply it to a state-of-art biomedical KG constructed from six datasets including biomedical relationships extracted from biomedical literature, which comprised 1,614,801 triples and 100,563 entities, including 30,006 diseases.

Results: Across five augmented architectures, KGiA improves generalizability by up to 24× in Mean Reciprocal Rank (MRR) and outperforms the state-of-the-art KG-based drug repurposing model by up to 32%. We applied KGiA in four case studies of diseases including Alzheimer's Disease and showed its promise in identifying novel repurposed candidate drugs.

Conclusion: We showed that leveraging counterfactual relationships derived from disease-specific graph structures to augment existing knowledge graphs improved performance in KG-based drug repurposing.

1. Introduction

Drug repurposing is the process of finding new applications for existing, approved drugs; it offers a time and cost-efficient approach to drug development [1,2]. Knowledge Graphs (KGs), defined as labeled, directed graphs with multiple, distinct edges between nodes [3], have recently been used for computational drug repurposing [4–6].

Previously published machine learning approaches for drug repurposing have predominantly been transductive models, which have an inherent limitation in requiring test nodes to be present in training datasets [7–9]. Several KG-based drug repurposing models are available, including KG-Predict [10], which infers new drug–disease interactions based on heterogeneous feature interactions between entity and relation embeddings in a biomedical knowledge graph.

Our work addresses this challenge through KGiA (Knowledge Graph-based drug repurposing with inductive and Augmentative capabilities), which can generalize to entirely unseen samples through inductive reasoning and disease-specific graph augmentation. This represents an advancement beyond current models that operate in the transductive space, such as NeoDTI [11], which integrates neighbor information

from heterogeneous networks, MolTrans [12], which uses transformer-based molecular interaction modeling, HyperAttentionDTI [13], which employs attention mechanisms for capturing complex molecular interactions, and EEG-DTI [14], which combines neuroimaging techniques with DTI analysis. Biomedical knowledge graphs (KGs) used in drug repurposing are inherently incomplete due to our limited knowledge of biomedical entities and their relationships [7–9]. KGiA addresses this limitation through targeted graph augmentation by identifying and leveraging disease-specific patterns to enhance the knowledge graph structure for more accurate drug repurposing predictions.

Domain-specific graph augmentation and inductive reasoning are key to advancing drug repurposing. We introduce KGiA which addresses both challenges: (1) identifying new drug repurposing opportunities through inductive reasoning without requiring additional training, and (2) enhancing prediction quality through disease-specific graph augmentation using equivariance preserving and counterfactual edges derived from high-quality biomedical datasets and literature.

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Statement of Significance

Problem	Knowledge graphs used in drug repurposing are inherently incomplete and struggle with efficient integration of new data while minimizing retraining time.
What is already known	Current link prediction models work well in transductive settings but struggle with inductive settings. Existing drug repurposing models rely primarily on transductive reasoning and lack domain-specific graph augmentation techniques.
Contributions of this paper	Introduces KGiA, an inductive KG augmentation method that leverages disease-specific topological structures and counterfactual relationships, to discover drug candidates.

2. Methods

2.1. System overview

A formal definition for knowledge graphs (KG) and problem formulation is presented in [Appendix A.1](#). Our approach supports multiple reasoning paradigms; it can perform fully-inductive reasoning (generalizing to entirely unseen data), semi-inductive reasoning (leveraging existing knowledge while handling new entities), and transductive reasoning (predicting new relationships between known entities) [15]. For drug repurposing applications, we demonstrate both transductive and semi-inductive reasoning since real-world drug discovery scenarios benefit from leveraging existing biomedical domain knowledge while simultaneously identifying novel drug–disease relationships. [Fig. 1](#) outlines our approach: First, a disease node is selected from the biomedical KG and its topological properties are analyzed (Section 3.3.1). Second, the dataset is split into transductive or semi-inductive settings (Section 2.4.2). Third, the training set is augmented with counterfactual edges based on topological similarities between node pairs, adding COUNTERFACTUAL_POSITIVE and COUNTERFACTUAL_NEGATIVE links (Section 2.3.1). Finally, a foundation model is fine-tuned on the augmented graph to rank drug candidates for the selected disease. The code is available at: <https://github.com/ceragoguztuzun/KGiA>.

The relation embedding process ([Appendix A.3](#), Algorithm 1) and the entity embedding process ([Appendix A.3](#), Algorithm 2) employ conditional representations meaning node embeddings are dynamically adjusted based on the query relation. Unlike standard GNN-based approaches (which use unconditional encoders) we explicitly condition node states on the query which makes the model more expressive [16, 17]. We provide more information on implementation in [Appendix A.3](#).

2.2. Foundation model

Foundation models are versatile AI systems engineered to generate a broad spectrum of outputs. These models can function independently or serve as the underlying framework for various other applications [18]. Galkin et al. [19] developed ULTRA, an approach for learning universal and transferable graph representations (see [Appendix A.4](#) for further details). Unlike standard methods that embed each relation independently, ULTRA explicitly represents relations as nodes within a separate “graph of relations,” where edges denote interactions or semantic connections between these relation types. By learning relation embeddings based on their interactions with other relations (i.e., relative representations) rather than relying on their fixed identities, ULTRA becomes invariant to permutations of relation and entity labels. Practically, this means the model can effectively generalize its learned knowledge to entirely new entities and relations, making it particularly suitable for inductive reasoning tasks. We utilize ULTRA as our state-of-the-art foundation model, adapting its application in the downstream task of identifying new drug repurposing opportunities.

2.3. Graph augmentation

We propose the following graph augmentation technique with two key strengths: (i) It incorporates both counterfactual relationships and domain-specific graph structures. (ii) It maintains the model’s invariant properties regarding double equivariance (Proof provided in [Appendix A.4, Theorem 1](#)).

2.3.1. Counterfactual relationships

Zhao et al. [20] defined counterfactual links to augment graph structure of training data in their rule-based approach. These links represent hypothetical outcomes between a pair of nodes that are not directly observable in the real-world data. The counterfactual links originate from the pivotal question in causal inference: “*Would the connection between the nodes persist if the observed graph structure were altered?*”. To understand this concept, consider how we might reason about cause and effect in drug discovery: rather than simply observing that a drug treats a disease, we ask what would happen if the drug’s topological properties in the graph were different. The process relies on causal models structured around three elements: the context, the manipulation, and the outcome [21,22].

Inspired by Zhao et al.’s work [20], for each drug–disease pair in our KG, we asked the question: “*Would Drug X treat Disease X if their selected topological graph structure became different from observation?*”. In our scenario of link prediction, we consider the data of a drug–disease node pair as the context, the characteristics of the graph’s selected topological structure as the manipulation, and the presence or absence of a TREAT link as the outcome.

For each drug–disease node pair $(dr_x, di_x) \in V \times V$ in our KG $G = (V, L, F)$, we find the most similar drug–disease node pair (dr_y, di_y) in G measured by $h(\cdot)$ which measures the Euclidian distance between a pair of nodes, with the opposite manipulation of the original node pair. The matrix M denotes manipulations on node pairs, with $M_{i,j} = 1$ if the drug dr_x and disease di_x share the same manipulation state and $M_{i,j} = 0$ otherwise. We use their outcome value to define counterfactual edges. These types of covariate matching are widely used to estimate manipulation effects [22,23].

We define two cases for the counterfactual link depending on the counterfactual answer, which are named to be distinguished from observed edges. This augmented graph represents a broader spectrum of possible interactions. COUNTERFACTUAL_POSITIVE links strengthen the hypothesis that dr_x could potentially treat di_x , even if they did not have the same topologic structure and COUNTERFACTUAL_NEGATIVE links are used to train our model to recognize not only existing but also potential non-effective treatments. The counterfactual edges are defined as the following:

$$\forall (dr_x, di_x) \in V \times V,$$

$$\begin{cases} dr_x \xrightarrow{\text{COUNTERFACTUAL_POSITIVE}} di_x & \text{if } dr_y \xrightarrow{\text{TREAT}} di_y, \\ dr_x \xrightarrow{\text{COUNTERFACTUAL_NEGATIVE}} di_x & \text{if } dr_y \not\xrightarrow{\text{TREAT}} di_y. \end{cases}$$

$$\text{where } (dr_y, di_y) = \underset{dr_y, di_y \in V}{\operatorname{argmin}} h((dr_x, di_x), (dr_y, di_y)) | M_{dr_y, di_y} = 1 - M_{dr_x, di_x}$$

The theoretical foundation for counterfactual edges lies in how they enhance graph reasoning while preserving double equivariance ([Appendix A.4, Theorem 1](#)). By identifying the most topologically similar pair (dr_y, di_y) with opposite manipulation state ($M_{dr_y, di_y} = 1 - M_{dr_x, di_x}$) for each drug–disease pair (dr_x, di_x) , we introduce structured learning signals based on therapeutic relationships. This approach ensures informative counterfactual pairs while maintaining structural contrasts that leads to improved reasoning capabilities as demonstrated in Section 3. Our approach systematically examines drug–disease pairs in the KG through counterfactual questioning. For each pair (dr_x, di_y) , we first identify their topological properties using various measures (Katz Centrality, Common Neighbors, Louvain, K-Core, Propagation) [24]. Their definitions are reproduced in [Appendix A.2](#). We then search

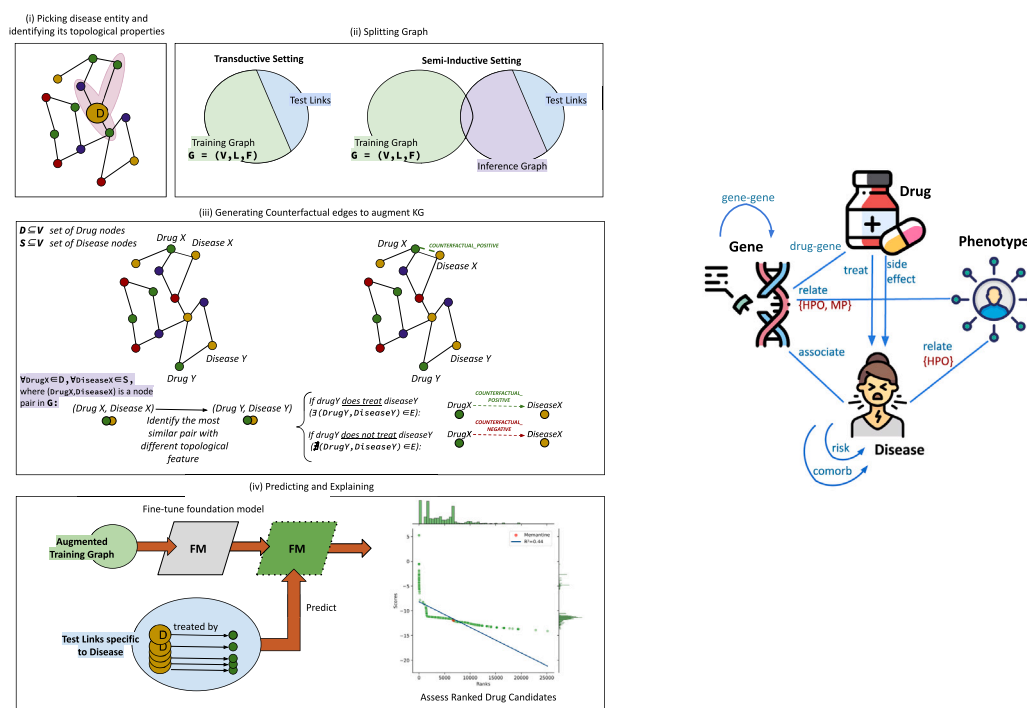


Fig. 1. (a) (i) Picking disease entity and identifying its topological properties, (ii) splitting graph [15], (iii) generating counterfactual edges to augment the knowledge graph, (iv) predicting drug candidates, explaining the intrinsic ranking mechanism of the model, analyzing drug mechanisms and evidence levels of drug candidates.

the embedding space to find another drug-disease pair (dr_a, di_b) with similar characteristics (close Euclidean distance) but opposite topological properties. This similarity in embedding space indicates shared fundamental properties, while their topological differences allow us to explore counterfactual scenarios.

Each topological property we measure reveals specific biological relationships. Consider Katz centrality: high centrality in a biomedical KG typically indicates that an entity has many well-studied interactions and plays a central role in multiple biological pathways. When creating counterfactual edges, we question *whether these biological implications matter for therapeutic effectiveness*.

A COUNTERFACTUAL_POSITIVE edge between dr_x (drug x) and di_y (disease y) indicates that, despite different centrality patterns from pair (dr_a, di_b) where dr_a treats di_b , therapeutic success remains possible. This suggests that *the biological interpretation of centrality (being well-studied and central to multiple pathways) might not be crucial for treatment effectiveness*. Conversely, a COUNTERFACTUAL_NEGATIVE edge suggests that *these biological implications of centrality patterns might be important*, as evidenced by the lack of therapeutic relationship between dr_a and di_b with opposite patterns.

This biological interpretation varies with each topological property [24]:

- *Katz centrality* questions the importance of global network influence
- *Common Neighbors* question the necessity of shared biological pathways
- *Louvain Communities* question the importance of functional module membership
- *K-Core* questions the significance of hierarchical positions in biological interactions
- *Propagation patterns* question the relevance of information flow through biological networks

Thus, rather than just examining abstract network properties we are questioning *whether the biological relationships these properties represent*

are necessary for therapeutic success. We provide a table of examples of augmentation edges for different manipulation choices and their interpretations in [Table A.4](#).

2.4. Experiment setting

To evaluate our model’s performance as a framework for KG reasoning in our experiments, we ask:

1. Does our model outperform a state-of-the-art drug repurposing model trained from scratch?
2. Does inductive generalization outperform transductive reasoning?
3. Does augmenting with a specific manipulation have any benefits?
4. (*Ablation Study*) Does introducing either counterfactual positive or negative edges provide any benefit?

2.4.1. Dataset

We constructed a KG by extracting interactions among genes, drugs, diseases, phenotypes, side effects, and symptoms from various high-quality public biomedical datasets (Fig. 1(b)). We sourced gene functions from the Mouse Genome Informatics (MGI) [25] and Human Phenotype Ontology (HPO) databases [26], and gene interactions from protein-protein interaction (PPI) networks [27]. Drug-side effect associations were obtained from the SIDER database [28]. Drug-gene associations were obtained from the DrugBank database [29]. Disease-phenotype associations and disease-gene interactions were obtained from the HPO and DisGeNET databases [30], respectively. The knowledge of disease comorbidities and drug-disease associations were extracted from 31,582,684 published biomedical research articles from Pubmed using the iterative pattern-based natural language processing techniques we previously developed [31–33]. The resulting KG comprises 100,563 entities across 4 types and 1,614,801 triplets across 10 edge types. Our KG is freely available at [ourGitHubrepository](#).

2.4.2. Setup

Data splitting is conducted in two frameworks: transductive and semi-inductive, both utilizing 70% of the total data for training. For both settings, we used the same training set. The data split is performed by randomly selecting triples from the graph. To address the high computational cost of data augmentation, we implemented hold-out validation. This approach avoids augmenting each fold in k-fold cross-validation individually while still ensuring robust evaluation through multiple runs with varied initializations (see Figure 6 in Appendix for details). In clinical research, model validation often involves testing on external datasets, which aligns more closely with the hold-out method, as suggested and employed by Eertink et al. [34], Liu et al. [35], Crowley et al. [36].

The semi-inductive setting features an inference graph with potentially new entities and relation types, combining validation and test data. This graph includes 212,266 unique triples not present in training data. The validation set encompasses 10,535 unseen entities, while the test set contains 17,926 unseen entities, with 7,591 entities unique to the test set. While our model can infer new relation types, no unseen relation types are present in the validation or test sets for this study. For the case study, we ensured that neither the training nor validation sets contained triples containing Alzheimer's Disease. Table A.7 details the number of augmentation relations for each manipulation strategy. Due to an imbalance between the counts of COUNTERFACTUAL_POSITIVE and COUNTERFACTUAL_NEGATIVE relations, we employed random downsampling on the majority group's triples to match with the numbers of the smaller group [37].

2.4.3. Baseline

We use KG-Predict [10] as our state-of-the-art KG-based drug repurposing computational framework baseline. Through the use of stacked CompGCN [38] layers and InteractE [39], entities and relations are embedded. As shown in their paper, they surpass DeepWalk [40], matrix factorization approaches, TransE [41], DistMult [42], ConvE [43], RotatE [44]. In addition to comparing generalizability, we compare the drugs repurposed by KG-Predict and our model in our case study.

2.4.4. Evaluation protocol

Inductive approaches inherently involve uncertainty when generalizing to unseen cases. In KGiA, this uncertainty stems from two sources: the model's need to generalize to new entities and the introduction of hypothetical counterfactual edges through graph augmentation. We address this through multiple evaluation metrics with hold-out validation, averaging across five training runs, and providing supporting evidence for predictions in our case study.

We evaluated KGiA using Mean Rank (MR), Mean Reciprocal Rank (MRR), Hits@k (k=1,3,10), AUROC, and AUPR. We acknowledge that each metric has limitations and advantages. For example, AUROC could overestimate the performance in imbalanced samples, while biologists may prioritize the performance of only top-ranked results.

As part of zero-shot inference, we use the inference graph and the test set of triples. For fine-tuning, we continue training the model on the training split, maintaining the checkpoint for the best MRR in the validation set. Whenever we run zero-shot experiments, we run them once since the results are deterministic, and we do not use augmentation as it only makes a difference in fine-tuned scenarios. We report an average of five runs for each fine-tuning run, and convert the average of MR into an integer by rounding up. A case study of repurposing for Alzheimer's Disease uses test split queries to perform link prediction to find drug candidates that have a high probability of having a TREAT relation with the disease node corresponding to Alzheimer's Disease (7845 such queries).

3. Results

3.1. Model performance

Table 1 showcases the models' performance across transductive and semi-inductive settings, focusing on their proficiency in prioritizing 'true' triples without pinpointing specific diseases or drug repurposing tasks. In the transductive scenario, the vanilla ULTRA foundation model outperforms KG-Predict, while augmented models demonstrate superior MRR and Hits@K metrics. Similarly, in the semi-inductive setting, augmented models exhibit comparable MR values but greatly improved MRR and Hits@K, highlighting our augmentation technique's effectiveness in improving link prediction tasks' generalizability and performance.

3.2. Impact of positive and negative counterfactual edges

In our ablation study, which aimed to analyze the impact of counterfactual positive and negative edges, we present the results in Table 2. In the transductive setting, we observed that adding only counterfactual positive or negative edges led to better performance than adding both together. However, in the semi-inductive setting, the combination of both edge types yielded better results.

This suggests that in the transductive setting, the model may be more sensitive to the introduction of additional information, and using either positive or negative counterfactual edges alone enhances performance without adding unnecessary complexity. In contrast, the semi-inductive setting, which lacks overlapping entities and relations, benefits more from the combined use of both edge types.

3.3. Case study

We applied our method to repurpose drugs for Alzheimer's Disease (AD). AD is a complex disease with mechanisms largely incompletely understood, but research has identified several key processes including amyloid plaques, tau tangles, neuroinflammation, and mitochondrial dysfunction [45–47]. There are several FDA-approved drugs targeting these mechanisms, however, currently, there are no cure or effective treatments for AD [48]. We have recently developed data-driven methods for AD drug repurposing [49–54]. We have recently developed KG-Predict for drug repurposing [10]. KG-Predict was evaluated on 30006 diseases using cross-validation and on AD as a case study. KGiA improved on KG-Predict using augmentation and inductive reasoning capabilities. In this study, we used AD as a case study to examine how KGiA improved on KG-Predict and other state-of-the-art methods for predicting repurposed candidate drugs for AD. Our study aimed to rank potential drug candidates $Drug \xrightarrow{\text{treats}} AD$, focusing on analyzing the top 5 ranked drug candidates.

3.3.1. Preliminary analysis of topological properties

To begin the case study with step (i) in Fig. 1, we analyze AD's topological properties. We plot the distributions of each manipulation choice, and we annotate the values of the entity corresponding to AD in Fig. 2 so that we can determine where AD falls within the distribution of the biomedical KG.

AD exhibits a Katz centrality score of 0.004, relatively high among the graph's nodes. By computing intra-community strength through the diagonal of the membership matrix M and inner-community strength by summing each row minus the diagonal, it is evident from Fig. 2(c) that AD's intra-community strength greatly surpasses its inter-community strength. This suggests AD may act as a bridge between communities. Further analysis reveals AD is part of the highest k-core of 144, and has a propagation-based integration level of 19520, typical for nodes within the KG.

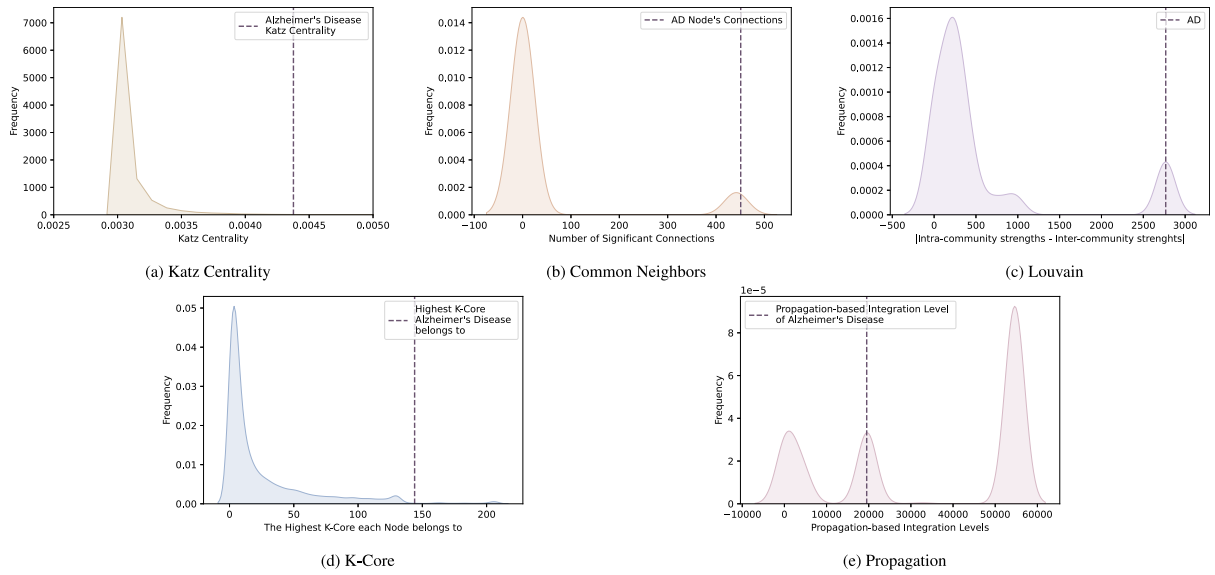


Fig. 2. Distributions of topological structures for augmentation manipulations. The node corresponding to Alzheimer's Disease is annotated.

Table 1

Evaluation metrics of models under Transductive and Semi-Inductive settings. We use KG-Predict as our primary benchmark, and non-augmented ULTRA models (zero-shot and fine-tuned) as secondary benchmarks. MR: Mean Rank; MRR: Mean Reciprocal Rank; Hits@K: Hit rate at top-K; AUROC: Area Under the Receiver Operating Characteristic Curve; AUPR: Area Under the Precision-Recall Curve.

Transductive							
Model	MR	MRR	Hits@1	Hits@3	Hits@10	AUROC	AUPR
KG-Predict	18 049	0.0118	0.0080	0.0111	0.0152	0.4533	0.0018
ULTRA 0-shot	49 748	0.0009	0.0002	0.0008	0.0018	0.9869	0.6857
ULTRA fine-tuned	3935	0.0178	0.0003	0.0185	0.0415	0.9787	0.9500
KGiA (Katz)	14 448	0.0302	0.0002	0.0328	0.0849	0.9893	0.9437
KGiA (Common Neighbors)	27 002	0.0035	0.0002	0.0042	0.0084	0.9795	0.5602
KGiA (Louvain)	20 295	0.0209	0.0001	0.0169	0.0487	0.9821	0.4109
KGiA (K-Core)	10 808	0.0227	0.0001	0.0197	0.0669	0.9889	0.3475
KGiA (Propagation)	4801	0.0338	0.0002	0.0271	0.0857	0.9968	0.6935
Semi-Inductive							
Model	MR	MRR	Hits@1	Hits@3	Hits@10	AUROC	AUPR
KG-Predict	–	–	–	–	–	–	–
ULTRA 0-shot	55	0.2958	0.1964	0.3449	0.4656	0.7622	0.2537
ULTRA fine-tuned	11	0.6351	0.4824	0.7404	0.9290	0.7443	0.2119
KGiA (Katz)	13	0.7456	0.6251	0.8512	0.9376	0.6108	0.1731
KGiA (Common Neighbors)	17	0.5856	0.4245	0.6964	0.8923	0.7656	0.1746
KGiA (Louvain)	23	0.8060	0.7437	0.8527	0.9108	0.6755	0.1735
KGiA (K-Core)	15	0.8380	0.7736	0.8939	0.9364	0.6001	0.1737
KGiA (Propagation)	20	0.8167	0.7444	0.8781	0.9280	0.7597	0.1839

Table 2

Ablation study on the effects of positive and negative counterfactual edge augmentations. Positive augmentation only introduces COUNTERFACTUAL_POSITIVE edges, and negative augmentation only introduces COUNTERFACTUAL_NEGATIVE edges. The reported values are the averages of all augmented models.

Augmentation	MR	MRR	Hits@10
Transductive			
Positive Aug	8286	0.0298	0.0805
Negative Aug	10 724	0.0269	0.0744
Both	15 471	0.0222	0.0589
Semi-Inductive			
Positive Aug	20	0.6666	0.8883
Negative Aug	14	0.7397	0.9051
Both	18	0.7584	0.9210

3.3.2. Evaluation

In Fig. 3, we compare the MR and MRR metrics of various models, with KG-Predict serving as our primary benchmark and non-augmented models (both zero-shot and fine-tuned) as secondary baselines. In the

transductive setting, Katz centrality, Common Neighbors, and Propagation augmented models outperform others in MR metrics (Fig. 3(a)). While KG-Predict outperforms non-augmented ULTRA in terms of MRR, augmented models, such as Katz, Common Neighbors, and Propagation, outperform (Fig. 3(b)). As for the semi-inductive setting, all augmented models, excluding Propagation, demonstrate superior performance regarding MR (Fig. 3(a)). As for MRR, Common Neighbors, Louvain, K-Core, and Propagation also exhibit strong performance (Fig. 3(b)).

Synthesizing these insights with Table 3's Hit@K values, the **Propagation augmented model** emerges as the top model for the transductive setting, and the **Common Neighbors augmented model** for semi-inductive setting.

The semi-inductive results for the AD case study differ significantly from the general case due to the strict formatting of triples, such as (drugX, treat, AD) (7845 such queries). This limited data variety increases sparsity, unlike the general setting, where flexible formats (as described in 2.4.2 Setup) allow richer data interactions and better model performance. The case study's narrower focus inherently presents challenges.

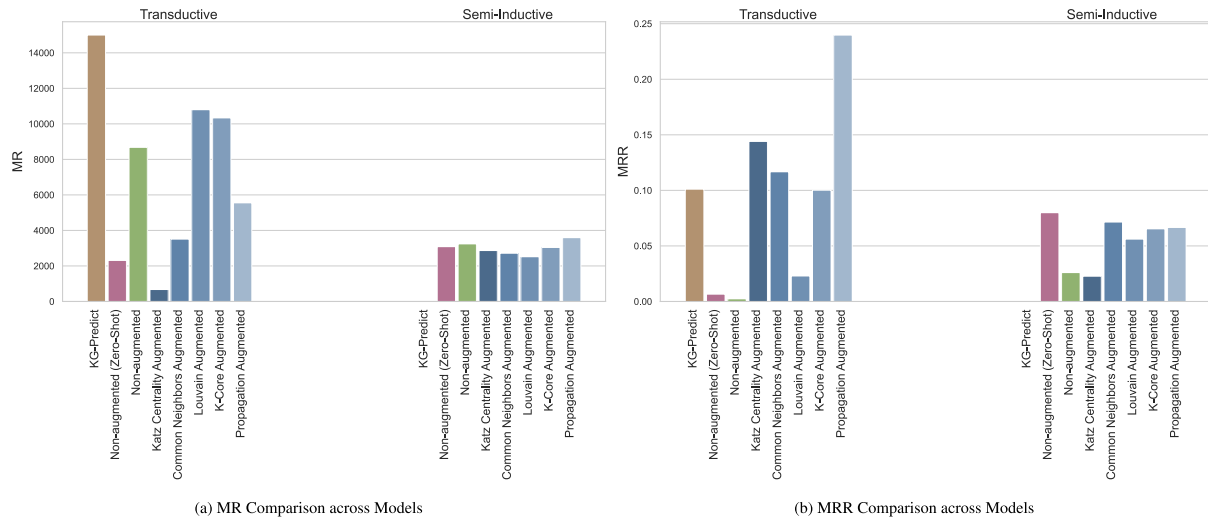


Fig. 3. Mean Rank (MR) and Mean Reciprocal Rank (MRR) values for the case study in both Transductive and Semi-Inductive settings.

Table 3

Hits@K performance for Alzheimer's Disease case study across models and settings (transductive vs semi-inductive)

Model	Transductive			Semi-inductive		
	Hits@1	Hits@3	Hits@10	Hits@1	Hits@3	Hits@10
KG-Predict	0	0	0	–	–	–
ULTRA 0-shot	0.00459	0.00459	0.00739	0.07559	0.07610	0.08107
ULTRA fine-tuned	0.00140	0.00166	0.00166	0.00025	0.03403	0.10121
KGiA (Katz Augmented)	0.03697	0.19401	0.31217	0.01651	0.01740	0.04442
KGiA (Common Neighbors Augmented)	0.04283	0.18917	0.20637	0.04066	0.11447	0.14800
KGiA (Louvain Augmented)	0.00688	0.03238	0.04576	0.05010	0.05048	0.05354
KGiA (K-Core Augmented)	0.09433	0.10057	0.11421	0.05851	0.06068	0.06386
KGiA (Propagation Augmented)	0.23429	0.23811	0.24551	0.06661	0.05163	0.07075

To further explore the generalizability of our method beyond AD, we conducted an exploratory post-hoc analysis on three additional diseases (Multiple Sclerosis, Cystic Fibrosis and Lung Cancer) with varying mechanisms and therapeutic landscapes. Full results and discussion are provided in [Appendix A.5](#).

3.3.3. Analysis of drug candidates

In [Fig. 4](#), we assess the models' internal consistency by assessing the ranks and scores given to each drug candidate. The curvature differences reflect how the model's certainty in its predictions changes across the ranking spectrum in each setting, and this is influenced by how the models were trained and how they generalize from seen to unseen data. In the transductive setting ([Fig. 4\(a\)](#)) where the model has access to all entities and relations during training, we observe a concave curve. In the semi-inductive setting ([Fig. 4\(b\)](#)), the model is based on a partial view of the network, leading to a steeper decline in confidence as the ranks increase.

When we look at both the Propagation augmented model and the Common Neighbors augmented model, the density plot shows a peak skewed to the left, suggesting a tendency to assign higher ranks to triples. Yet, the scatters corresponding to ranked triples convey that many predictions are tied to negative logit scores, highlighting a lack of confidence in those outcomes. On the other hand, the Non-Augmented models maintain a moderate level of confidence throughout their rankings.

We listed the drug candidates for each model in [Fig. 5](#), as well as each drug's evidence level and mechanism category. [Fig. 5](#) is a high-level summary of the details presented in [Table A.5](#) (transductive setting) and [Table A.6](#) (semi-inductive setting) in Appendices.

In the transductive setting, treatments for neurological and psychiatric disorders are prioritized, leading to a range of drugs with

varying evidence levels for repurposing in AD. The non-augmented model presents the same drug candidates as the baseline model KG-Predict, except in a different order. In contrast, our augmented model recommends additional candidates that show weak evidence of potential repurposing for AD, differing from both the KG-Predict baseline and the non-augmented model.

In the semi-inductive setting, as the model was not trained on the complete set of entities and relations, the drug candidates are more diverse than in the transductive setting. We find that the top two drug candidates (Aducanumab, Ipidacrine) are identical for both the non-augmented and augmented models, exhibiting strong evidence in the mechanisms of hormonal immunomodulatory agents and treatments for neurological and psychiatric disorders ([Table A.5](#)). The augmented model has identified a greater number of drugs with no existing references, indicating these require further investigation.

4. Discussion and conclusion

Addressing incompleteness of KGs requires efficiently integrating new data with minimal re-training. To meet this need, we introduce KGiA, a domain-specific augmentation approach enabling semi-inductive reasoning. [Table 1](#) illustrates that inductive reasoning significantly improves generalization metrics (MRR, Hits@K) compared to transductive models, primarily due to the model's capacity to handle heterogeneous connectivity.

Leveraging counterfactual relationships derived from disease-specific topologies dramatically improved metrics, tripling the MRR and boosting Hits@K. This augmentation approach effectively leverages domain knowledge and increases the diversity of drug candidates. For Alzheimer's Disease (AD), augmented models predicted drugs with 5

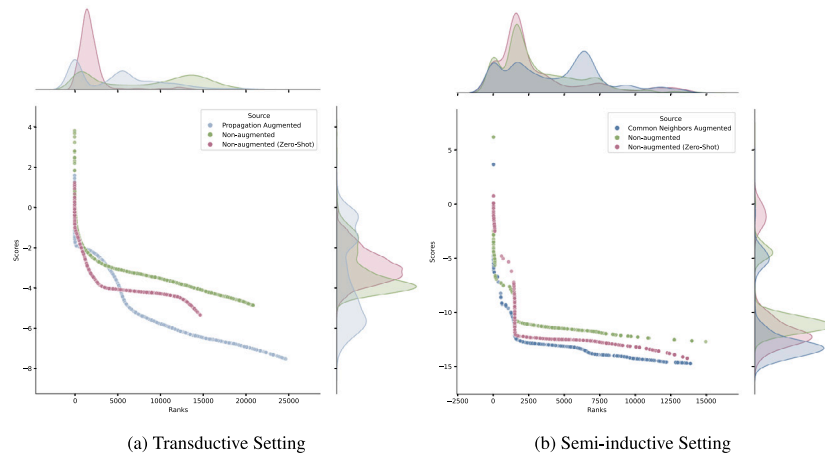


Fig. 4. Rankings and scores as predicted by models, assessing internal consistency within models based on the case study. The best augmented models (Propagation and Common Neighbors) and non-augmented baselines are shown. Each scatter indicates a drug candidate ranked by the model, with logit scores.

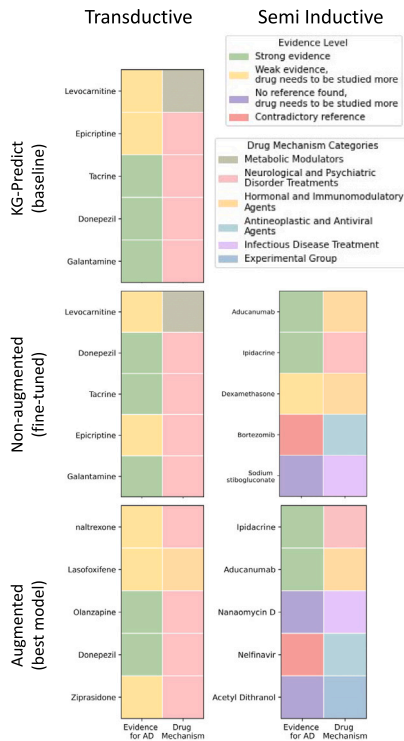


Fig. 5. Top 5 Drug Candidates proposed across models. This figure is a high-level summary of the evidences and drug mechanisms provided in Tables A.5 and A.6. Best models for the augmented setting are Propagation augmented model for the transductive setting, and Common Neighbors augmented model for the semi-inductive setting.

distinct mechanisms of action compared to 2 with baseline models (Fig. 5), which could help personalized prescriptions.

Note that low MRR and Hits@K metrics are typical in sparse biomedical KGs due to few true positives within large candidate pools. ULTRA fine-tuned exhibited strong AUROC and AUPR scores (Table 1) which highlight the model's robust probabilistic ranking despite limited precision at top-k positions.

The effectiveness of augmentation strategies depends on the disease entity's position within the topological structure distribution. Centrally located entities provide more informative and robust augmentation results, while extreme positions reduce consistency. For AD, Propagation

and Common Neighbors methods (centrally positioned methods, Fig. 2) were most effective for transductive and semi-inductive reasoning, respectively.

Scalability remains a challenge due to increased computational demands from identifying counterfactual edges. Inductive reasoning, while powerful, inherently introduces uncertainty, which we mitigate via hold-out validation, multiple performance metrics, and averaging. Our evaluation framework relies on standard ranking and classification metrics each with known limitations regarding sparsity, imbalance, and the biological relevance of predictions. To partially address this, our AD case study included domain-expert validation of predicted drug candidates (Fig. 5, Tables A.5, A.6).

ULTRA's low Hits@1 reflects its difficulty learning from small datasets, yet its foundational role in enabling inductive reasoning and improved generalization through double equivariance justifies our selection. Future research should explicitly evaluate double equivariance's contribution, enhance scalability, and validate model outputs across a wider set of diseases. Evaluation frameworks should incorporate a balanced set of ranking-based and probabilistic metrics alongside biological plausibility checks to provide a performance assessment. As a limitation, not updating the underlying knowledge graph with the most recent databases may constrain model performance and generalizability, since newer resources such as OnSIDES [55] achieve higher precision and recall than SIDER 4.1 [28]. To enable a fair comparison with KG-Predict [10], which also uses SIDER, we applied the same dataset across all models; future work will integrate OnSIDES and other up-to-date sources to evaluate how sensitive our methods are to the choice of knowledge base. This study lays the groundwork for sophisticated domain-specific graph augmentations that mirror observable biological structures.

CRediT authorship contribution statement

Çerağ Oğuztüzün: Writing – original draft, Visualization, Software, Methodology, Formal analysis, Conceptualization. **Zhenxiang Gao:** Writing – review & editing, Data curation. **Hui Li:** Validation. **Rong Xu:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Appendices

A.1. Problem formulation

A.1.1. Knowledge graph definition

We define KGs as labeled, directed graphs with multiple, distinct edges between nodes [3]. Formally, a KG is represented as $KG = (V, L, F)$, where V is a set of vertices, each symbolizing an entity. L is a collection of labels, each indicating a type of relation. $F \subseteq V \times L \times V$ is a set of directed edges that depict relationships as triples $\langle h, r, t \rangle$, with h as the head entity, r as the relation label, and t as the tail entity.

A.1.2. Link prediction in graphs

The link prediction task in the context of KG reasoning answers queries $\langle h, r, ? \rangle$. The problem involves estimating the likelihood of a specific link $\langle h, r, t \rangle$ being true but not observed in the current graph structure. Given a query $\langle h, r, ? \rangle$, our goal is to rank all nodes $v \in V$ in the inference graph, denoted as $KG = (V, L, F)$, such that the probability $P(t = v \mid h, r)$ of correctly identifying a true tail entity t is maximized. For drug repurposing, the link prediction problem can be specialized to predict new $Drug \xrightarrow{\text{treats}} Disease$ triples [56]. This involves estimating the likelihood of a given drug being effective in treating a disease that it is not currently known or approved to treat, based on the knowledge encoded in the KG [57].

In the *transductive link prediction* setting, the model is trained and inferred on the same graph $G_{\text{train}} = G_{\text{inf}}$. The model is expected to predict new links between entities that were already part of the graph during the training phase but does not generalize to unseen entities or relations. Where $G_{\text{train}} = (V_{\text{train}}, L_{\text{train}}, F_{\text{train}})$ and $G_{\text{inf}} = (V_{\text{inf}}, L_{\text{inf}}, F_{\text{inf}})$: $V_{\text{train}} = V_{\text{inf}}$, $L_{\text{train}} = L_{\text{inf}}$, $F_{\text{train}} \subseteq F_{\text{inf}}$.

In the *semi-inductive setting*, the model is trained on a graph where it may not have seen all the entities (V) and relations (L) during training, but it is expected to generalize to unseen entities ($V_{\text{inf}} \setminus V_{\text{train}}$) and relations ($L_{\text{inf}} \setminus L_{\text{train}}$) in the same graph during inference [58]. This means there is a partial overlap between the training and inference graphs, particularly with some new entities or relationships being introduced at inference time that were not present during training such that: $V_{\text{train}} \subseteq V_{\text{inf}}$, $L_{\text{train}} \subseteq L_{\text{inf}}$, $F_{\text{train}} \subseteq F_{\text{inf}}$.

A.2. Manipulation definitions

We experimented with the following manipulations [24], where A is the adjacency matrix for graph G :

- **Katz Centrality:** $M = ((I - \beta A)^{-1} - I) \geq \theta$, θ is a threshold value and $\beta = \min\left(\frac{1}{\lambda_{\max}}, 0.003\right)$
- **Common Neighbors:** $M = ((A + A^2 + A^3 > 0) \times (A + A^2 + A^3 > 0)^T \geq k)$, k is a threshold value.
- **Louvain:** $M = (LL^T)$, L is the membership matrix derived from applying Louvain to A .
- **K-Core:** $M = (KK^T)$, K is the membership matrix derived from the k-core labels of the nodes in A .

- **Propagation:** $M = (PP^T)$, P is the membership matrix derived from the propagation clustering community labels of the nodes in A .

Comparing M_{dr_x, di_x} with M_{dr_y, di_y} evaluates the structural similarity between drug–disease pairs through different topological lenses: Global network influence (Katz Centrality), Local connectivity patterns (Common Neighbors), community membership (Louvain), network density positioning (K-Core), information flow patterns (Propagation).

As each captures distinct network characteristics, this explains the varying performance of different augmentation methods for Alzheimer's Disease. The condition $M_{dr_y, di_y} = 1 - M_{dr_x, di_x}$ identifies pairs with opposite structural properties, enabling counterfactual reasoning: "Would Drug X treat Disease X if its network position had opposite properties?"

A.3. Embedding process

The relation embedding process is shown in Algorithm 1, and the entity embedding and link prediction process is shown in Algorithm 2. The foundation model captures interactions by creating a graph of relations $G_r = (R, R_{\text{fund}}, E_r)$ to map out how different types of relations interact. In G_r , all $r \in R$ are unique relation types in the original graph G , and two nodes are linked if the corresponding types of relations in G share a common entity. $R_{\text{fund}} = \{\text{tail-to-tail edges, head-to-head edges, head-to-tail edges, tail-to-head edges}\}$ which are the four types of links in G_r that describe the basic topological structure of a directed graph.

Specifically, in our graph of relations, G_r , each node represents a unique relation type from the original graph, G . Two nodes in G_r are connected if their corresponding relation types in G share a common node. Depending on the directionality of these connections in G , the connections in G_r are classified as either tail-to-tail, head-to-head, head-to-tail, or tail-to-head.

Embeddings are obtained using conditional representations [16], which are demonstrably more expressive and practical than unconditional GNN encoders [17,59]. By doing so, node representations are conditioned on query relations.

In Algorithm 1 and 2, **AGG** is a summation aggregation function and **MSG** is a multiplication messaging function.

Algorithm 1 Relation Graph Embedding

```

1: Input: Original graph  $G$ , Query  $(h, q, ?)$ , Dimension  $d$ 
2: Output:  $d$ -dimensional node representations  $N_q$  of  $G_r$ 

3: function COMPUTErepr( $q, G_r, d$ )
4:   for each node  $u$  in  $G_r$  do
5:      $N_q[u] \leftarrow (u == q) ? \mathbf{1}^d : \mathbf{0}^d$ 
6:   return  $N_q$ 

7: function MESSAGEPASSING( $G_r, N_q$ )
8:    $t \leftarrow 0$ 
9:   repeat
10:    for each node  $v$  in  $G_r$  do
11:       $msgs \leftarrow \{\text{MSG}(h_w[t], r) \mid w \in \text{neighbors}(v), r \in R_{\text{fund}}\}$ 
12:       $agg \leftarrow \text{AGG}(msgs)$ 
13:       $h_v[t+1] \leftarrow \text{UPDATE}(h_v[t], agg)$ 
14:     $t \leftarrow t + 1$ 
15:   until convergence
16:   return  $h_v[t]$ 

17:  $G_r \leftarrow \text{ConstructGr}(G)$ 
18:  $N_q \leftarrow \text{ComputeRepr}(q, G_r, d)$ 
19:  $N_q \leftarrow \text{MessagePassing}(G_r, N_q)$ 

```

Table A.4

Examples of counterfactual relationships and their interpretations.

Topology type	Drug–disease pair (augmentation edge)	Counterfactual question	Biological interpretation
Katz Centrality	Tibolone(DB09070) $\xrightarrow{\text{COUNTERFACTUAL_POSITIVE}}$ Ovarian Carcinoma(C0677886)	What if being globally influential and connected in biological pathways is not necessary for treating ovarian cancer?	Since similar drug–disease pairs work despite different centralities, tibolone might treat ovarian carcinoma regardless of its network centrality
Common Neighbors	Propoxycaïne(DB09342) $\xrightarrow{\text{COUNTERFACTUAL_NEGATIVE}}$ Alexander Disease(C0270726)	What if sharing local biological pathway connections with known treatments for white matter disorders is not necessary?	Since similar pairs do not work without shared pathways, propoxycaïne might not treat Alexander disease due to insufficient pathway overlap
Louvain Community	Berberine(DB04115) $\xrightarrow{\text{COUNTERFACTUAL_POSITIVE}}$ Myocardial Infarction(C0027051)	What if belonging to the same functional module as known cardiac treatments is not necessary?	Since similar pairs work across different biological modules, berberine might treat myocardial infarction despite being in a different functional module

Algorithm 2 Entity Embedding and Link Prediction

```

1: Input: Graph  $G$ , Query  $(h, q, ?)$ , Conditional relation representations  $R_q$ , Dimension  $d$ 
2: Output: Logits  $p(h, q, v)$  for each node  $v$  in  $G$ 

3: function INITGRAPH( $G, h, q, R_q, d$ )
4:    $h_0 \leftarrow \text{Vector}(|G|, d, 0)$   $\triangleright$  Zero vector for each node
5:    $h_0[h] \leftarrow R_q[q]$   $\triangleright$  Set query vector for head node
6:   return  $h_0$ 

7: function MESSAGEPASSING( $G, h_0, R_q, d$ )
8:    $t \leftarrow 0$ 
9:    $R_t \leftarrow \text{TransRelations}(R_q)$   $\triangleright$  Apply 2-layer MLP
10:  repeat
11:    for each node  $v$  in  $G$  do
12:       $\text{msgs} \leftarrow \{\text{MSG}(h_w[t], R_t[r]) \mid w \in \text{neighbors}(v)\}$ 
13:       $\text{agg} \leftarrow \text{AGG}(\text{msgs})$ 
14:       $h_v[t+1] \leftarrow \text{UPDATE}(h_v[t], \text{agg})$ 
15:     $t \leftarrow t+1$ 
16:  until convergence
17:  return  $h_v[t]$ 

18:  $h_0 \leftarrow \text{InitGraph}(G, h, q, R_q, d)$ 
19:  $h_v \leftarrow \text{MessagePassing}(G, h_0, R_q, d)$ 
20:  $\text{logits} \leftarrow \text{ComputeLogits}(h_v)$ 

```

A.4. Preserving equivariance**A.4.1. Invariance and equivariance**

In inductive inference, neural networks are trained to discern and generalize invariant features within a graph to accommodate novel data. Xu et al. [60] and Morris et al. [61] have revealed a notable parallel between the Weisfeiler–Lehman (WL) test and the operations of graph message passing neural networks, which facilitate convolution-like functionalities on graphs. A standard GNN updates the hidden state $h_v^{(k)}$ of a node v at layer k as: $h_v^{(k)} = \phi(h_v^{(k-1)}, f(\{h_u^{(k-1)} : u \in N(v)\}))$ [62], where ϕ denotes a transformation operation, f denotes an aggregation operation, $N(v)$ denote the neighboring nodes from previous layer $k-1$. Whereas, a 1-WL test iteration represents the label $l_v^{(k)}$ of node v as: $l_v^{(k)} = g(l_v^{(k-1)}, \text{sort}\{l_u^{(k-1)} : u \in N(v)\})$ [60], where g is an injective function. Xu et al. [60] show by induction that (Theorem 3 in their paper) there always exists an injective function φ that $h_v^{(k)} = \varphi(l_v^{(k)})$, which show that WL’s 1-dimensional form 1-WL is analogous to GNN’s neighbor aggregation. Showing the existence of the injective function φ is stating that for any given unique label $l_v^{(k)}$, there is a unique embedding $h_v^{(k)}$ in the GNN. It implies that these embeddings

are independent of node IDs, similar to the WL test’s ability to create unique labels for nodes based on their structure, rather than arbitrary properties like node IDs. Hence, these studies have underscored the benefits for inductive graph tasks from assuming node IDs are arbitrary, suggesting that models should maintain consistency despite permutations of node IDs. This concept, known as *permutation equivariance*, implies that the network does not rely on node IDs for embedding but rather employs the graph’s structural topology for representation [63].

Extending this to multi-relational graphs such as KGs, the notion of learnable invariances extends to the joint permutations of node IDs and relation IDs, a concept referred to as *double permutation equivariance* or *double equivariance* defined by Gao et al. [59]. If you consider a graph that is processed by an RGCN (Relational Graph Convolutional Network) [64], the goal is for the RGCN to learn representations of nodes and relations from their structure rather than their labels or order. Given we have a permutation operation π , which shuffles the order of nodes and the types of relations between them. Permutation invariance in multi-relational graphs like RGCNs can be captured by demonstrating that the aggregation function f we defined previously is permutation-invariant concerning the representations of neighboring nodes of v through relation type r as $(N^r(v))$, then permutation invariance is established which can be defined as the aggregation function f_r :

$$f_r(\{h_u^{(k-1)}, w_r : u \in N^r(v)\}) = f_r(\{h_{\pi(u)}^{(k-1)}, w_r : \pi(u) \in \pi(N^r(v))\}) \quad (\text{A.1})$$

Theorem 1 (Theorem 1). *Graphs with counterfactual edge augmentation preserve double equivariance.*

$$f_r(\{h_u^{(k-1)}, w_r : u \in N_{cf}^r(v)\}) = f_r(\{h_{\pi(u)}^{(k-1)}, w_r : \pi(u) \in \pi(N_{cf}^r(v))\}) \quad (\text{A.2})$$

Proof. Let $G = (V, L, F)$ be the original KG, and $G' = (V, L \cup L_{cf}, F \cup F_{cf})$ represent the graph augmented with counterfactual edges. N_{cf}^r is the neighborhood of node v in G' . The aggregation function for relation r at layer k is denoted as f_r , which aggregates features from the neighborhood $N_{cf}^r(v)$ of a node v through relation r . Initially defined in Equation 1, the relation-based aggregation function f_r is permutation-invariant (see Fig. A.6). Since the addition of counterfactual edges L_{cf}, F_{cf} to the graph G to form G' simply extends the neighborhood $N_{cf}^r(v)$ without altering the inherent permutation-invariant property of f_r , the aggregation function will process $N_{cf}^r(v)$ and $\pi(N_{cf}^r(v))$ identically. Therefore, f_r preserves double equivariance in the presence of counterfactual edges, as the permutation of node IDs does not affect the outcome of the aggregation function. $\square$

The full versions of the tables and the tables for other models’ drug candidates are presented in our [GitHub repository](#).

Table A.5

Evidence and mechanism of the Top 3 drug candidates for Alzheimer's Disease, predicted by transductive model.

Drug candidates	Evidence for AD	Drug mechanism
KG-Predict		
Levocarnitine	No clinical trial for levocarnitine itself, but clinical trial NCT02955706 for its derivative Acetyl-L-carnitine (L-carnitine is a synonym for levocarnitine)	Prevent and treat a lack of carnitine in patients with kidney disease on dialysis. A carrier of long chain fatty acids to mitochondria for energy.
Epicriptine	No reference, but with indication in Alzheimer's disease on Drug Central website and its analog dihydroergocristine is proved to delay progression of Alzheimer's disease [65]	Nootropic to treat signs and symptoms of an idiopathic decline in mental capacity. Mainly due to the agonistic activity on dopamine receptor
Tacrine	First drug approved for Alzheimer's disease	Treat AD. A powerful acetylcholinesterase inhibitor (AChE)
Non-augmented		
Levocarnitine	No clinical trial for levocarnitine itself, but clinical trial NCT02955706 for its derivative Acetyl-L-carnitine (L-carnitine is a synonym for levocarnitine)	Prevent and treat a lack of carnitine in patients with kidney disease on dialysis. A carrier of long chain fatty acids to mitochondria for energy.
Donepezil	NCT02787746(completed), NCT04661280(recruiting)	Treat AD. Inhibiting the acetylcholinesterase enzyme
Tacrine	First drug approved for Alzheimer's disease	Treat AD. A powerful acetylcholinesterase inhibitor (AChE)
KGiA (Propagation Aug)		
naltrexone	Serby et al. [66] and Hyman et al. [67] (but show no significant improvement)	Treat alcohol dependence and block the effects of exogenously administered opioids. It is a pure opiate antagonist and has little or no agonist activity.
Lasofixifene	Selective estrogen receptor modulators as brain therapeutic agents (further study is needed)	An estrogenic and antiestrogenic drug used in both vitro and in vivo to treat osteoporosis and vaginal atrophy.
Olanzapine	NCT00015548(completed, olanzapine is used in combination with other two drugs)	The drug is used to treat schizophrenia, bipolar disorder, and agitation associated with these disorders. It inhibits multiple neuronal receptors, including D1, D2, D3, and D4.

Table A.6

Evidence and mechanism of the Top 3 drug candidates for Alzheimer's Disease, predicted by semi-inductive model.

Drug candidates	Evidence for AD	Drug mechanism
Non-augmented		
Aducanumab	NCT04241068 (active, not recruiting), NCT02477800(terminated with results)	Treat AD. It is a human immunoglobulin gamma 1 (IgG1) monoclonal antibody which selectively targets and binds aggregated soluble oligomers and insoluble fibril conformations of A β plaques.
Ipidacrine	Also named Neiromidin or Neuromidin(translated from Russian) [68].	Treat memory disorders of different origins. A reversible acetylcholinesterase inhibitor
Dexamethasone	Alisky et al. [69] (It is a case report and hypothesis, needs further validation)	Inflammatory disorders such as asthma, endocrine disorders, and rheumatoid arthritis are treated with glucocorticoids that inhibit pro-inflammatory signals and promote anti-inflammatory ones.
KGiA (Common Neighbors Aug)		
Ipidacrine	Also named Neiromidin or Neuromidin(translated from Russian) [68].	Treat memory disorders of different origins. A reversible acetylcholinesterase inhibitor
Aducanumab	NCT04241068 (active, not recruiting), NCT02477800(terminated with results)	Treat AD. It is a human IgG1 monoclonal antibody which selectively targets and binds aggregated soluble oligomers and insoluble fibril conformations of A β plaques.
Nanaomycin D	No reference	An enantiomer of the antibiotic kalafungin, is a natural product found in Streptomyces and Streptomyces rosa.

Table A.7

Number of counterfactual edges found in the knowledge graph.

Counterfactual negative edges		Counterfactual positive edges	
Method	Count	Method	Count
Katz Centrality	3.15×10^7	Propagation	3.07×10^7
Louvain	2.80×10^7	Common Neighbors	3.06×10^7
K-Core	2.62×10^7	K-Core	5.49×10^6
Common Neighbors	1.21×10^6	Louvain	3.73×10^6
Propagation	9.95×10^5	Katz Centrality	2.12×10^5

A.5. Exploratory case studies on additional diseases

Beyond our primary case study on Alzheimer's Disease (AD), we conducted an exploratory evaluation on three additional diseases with

diverse pathological profiles: Multiple Sclerosis (MS), Cystic Fibrosis (CF), and Lung Cancer (LC). MS is an autoimmune neurodegenerative disorder with limited treatment options; CF is a genetic disease with well-understood mechanisms and available disease-modifying therapies; and LC represents a complex malignancy with heterogeneous treatment responses.

This broader evaluation serves to probe KGiA's applicability across a range of clinical contexts and treatment landscapes. As shown in Fig. A.7, augmentation strategies yielded performance improvements in MS and CF, while LC benefited less from augmentation, which highlights that disease-specific characteristics may influence which augmentation strategy is most effective. These exploratory results suggest that model generalization may benefit from tailoring augmentation approaches to the biological and clinical nature of the target disease.

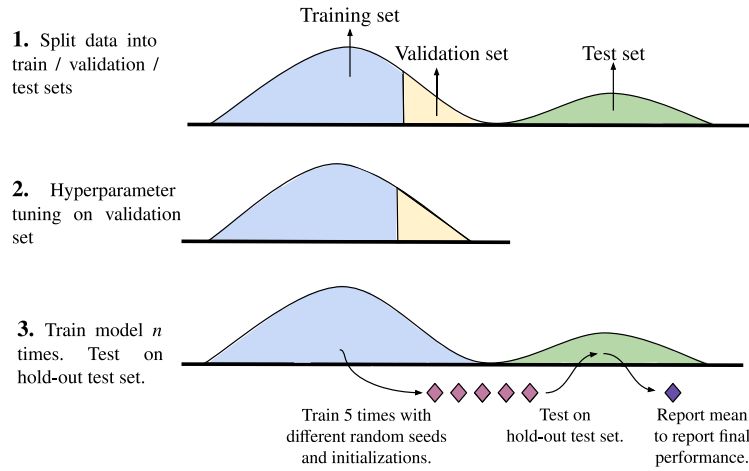


Fig. A.6. Overview of our evaluation process: (i) The dataset was divided into a training set and a fixed test set. (ii) Hyperparameter tuning was conducted using the validation set. (iii) Each model was trained five times with different random seeds and initializations, while evaluation was consistently carried out on the same fixed test set. Performance metrics were collected from each of the five runs, and the average of these metrics was reported as the final performance result.

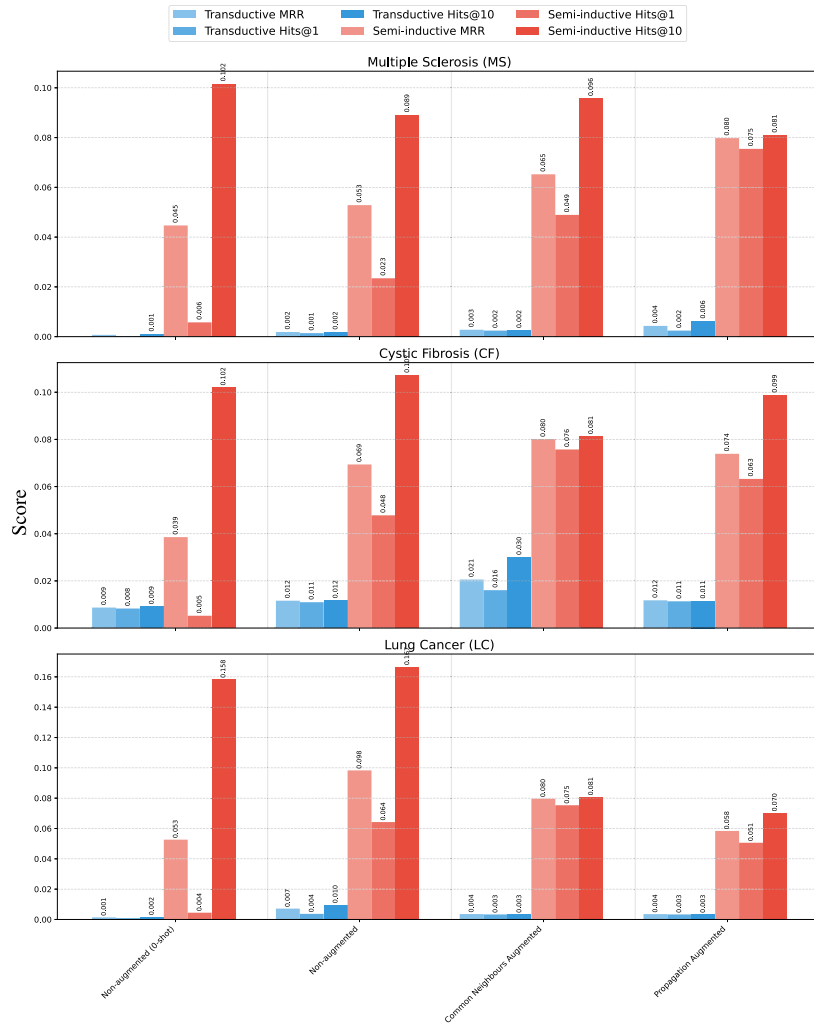


Fig. A.7. Comparative analysis of different model architectures across transductive and semi-inductive settings for Multiple Sclerosis (MS), Cystic Fibrosis (CF), and Lung Cancer (LC). This analysis complements the AD case study in Fig. 3 and Table 3. Different diseases benefit from different augmentation strategies: AD and MS from propagation methods, CF from common neighbors, and LC from non-augmented models. These findings support the need for disease-aware augmentation strategies in drug repurposing tasks.

Appendix B. Supplementary data

Supplementary materials are provided in: <https://github.com/ceraoguztuzun/KGIA>.

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